

Threshold Entry in a Fixed-Prize Contest

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1 Introduction

Table 1: Headline claims, each traced to the model rows it rests on.

ID	Headline	Model rows	Scope note
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2 Model

Table 2: Results of the model, each with its statement and the Lean theorems that check it.

ID	Claim	Statement	Lean	Proof
M1	Representation (P1).	$\Delta(m) = V \int \phi dH_m = V \mathbb{E}[\phi(M_m)]$, with $M_m \sim H_m$.	EntryContestP1P2.two_max_sum, EntryContestP1P2.p1_representation, EntryContestP1P2.Delta_eq_expectation	p. 2
M2	Dominance (P2).	For $s \geq 0$ and $\mu \leq 1$, $F \leq G$ and $\Delta(m) \geq 0$. For $\mu < 1$ and $\Pr(s > 0) > 0$, $F < G$ on some $[q, u)$ with $q < u$.	EntryContestP1P2.p2_fosd, EntryContestP1P2.p2_strict_interval, EntryContestP1P2.Delta_nonneg_from_primitives	p. 3
M3	Step identity (P3).	For $m + 2 \leq Q$, $\Delta(m+1) - \Delta(m) = -\frac{V}{2} \int \phi^2 dK_m$.	EntryContestStep.three_max_sum_real, EntryContestStep.step_identity, EntryContestModel.Delta_step	p. 3
M4	Monotonicity.	$\Delta(m+1) \leq \Delta(m)$ for $V \geq 0$, with $<$ when $V > 0$ and $\int \phi^2 dK_m > 0$.	EntryContestModel.Delta_step_nonpos, EntryContestModel.DeltaUpTo_step, EntryContestEq.Delta_step_strict	p. 3
M5	Threshold entry (P5).	For $\kappa_0 \leq \dots \leq \kappa_{Q-1}$ and Δ non-increasing, $\kappa_j \leq \Delta(j)$ and $i \leq j$ give $\kappa_i \leq \Delta(i)$.	EntryContest.single_crossing, EntryContestModel.p5_threshold_from_primitives	p. 4

ID	Claim	Statement	Lean	Proof
M6	Count invariance.	Every equilibrium S , with $\kappa_i \leq \Delta(S - 1)$ for $i \in S$ and $\Delta(S) < \kappa_j$ for $j \notin S$, has $ S = k^*$.	EntryContestEq.count_invariance_fin, EntryContestMM.exists_mem_ge, EntryContestMM.exists_not_mem_le	p. 4
M7	Identities pinned.	If $\Delta(k^* - 1) < \kappa_{k^*}$, the assortative set $\{0, \dots, k^* - 1\}$ is the only equilibrium.	EntryContest.members_below_kstar, EntryContestEq.assortative_unique	p. 4
M8	Assortative selection (N4).	$\{0, \dots, k^* - 1\}$ is an equilibrium; $\sum_{i < k} \kappa_i \leq \sum_{i \in S} \kappa_i$ for $ S = k$; with identical score laws, payoff sums of equal-size equilibria differ by their cost sums.	EntryContestEq.assortative_is_equilibrium, EntryContestEq.assortative_min_cost, EntryContestEq.prize_term_anonymous, EntryContestEq.payoff_gap_anonymous	p. 4

3 Literature

Table 3: Papers the model or the conclusions depend on, each with a verbatim quote.

ID	Paper	What we rely on	Quote	Page	Model rows
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4 Conclusions

Table 4: The novel claims, each linked to the model and literature rows it rests on.

ID	Statement	Rests on
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A Proofs

Notation. Challenger i scores $\sigma_i = \mu r_i + (1 - \mu)s_i e_i$, with r_i and $s_i \geq 0$ drawn independently of wealth. F and G are the score CDFs of an entrant and of an outsider, and C is the incumbent's, each without atoms. With m other entrants among the Q challengers,

$$H_m = C F^m G^{Q-1-m}, \quad K_m = C F^m G^{Q-2-m}, \quad \varphi = G - F,$$

$$\Delta(m) = V \left(\int H_m dF - \int H_m dG \right), \quad \kappa(w) = u(w) - u(w - c).$$

Challengers are indexed $0, \dots, Q - 1$ in increasing order of cost, $\kappa_0 \leq \dots \leq \kappa_{Q-1}$, which is decreasing order of wealth. Index j enters when $\kappa_j \leq \Delta(j)$, and k^* satisfies $\kappa_j \leq \Delta(j)$ for $j < k^*$ and $\Delta(k^*) < \kappa_{k^*}$ when $k^* < Q$.

Proof of M1

Lean: EntryContestP1P2.two_max_sum, EntryContestP1P2.p1_representation, EntryContestP1P2.Delta_eq_expectation.

For independent $X \sim A$ and $Y \sim B$ with B atomless, $\Pr(X = Y) = 0$, so

$$\int B dA + \int A dB = \Pr(Y \leq X) + \Pr(X \leq Y) = 1.$$

With $(A, B) = (F, H_m)$ and $(A, B) = (G, H_m)$,

$$\int H_m dF = 1 - \int F dH_m, \quad \int H_m dG = 1 - \int G dH_m,$$

so

$$\frac{\Delta(m)}{V} = \int (G - F) dH_m = \int \phi dH_m = \mathbb{E}[\phi(M_m)].$$

□

Proof of M2

Lean: EntryContestP1P2.p2_fosd, EntryContestP1P2.p2_strict_interval, EntryContestP1P2.Delta_nonneg_from_primitives.

Let $X = \mu r + (1 - \mu)s$ and $Y = \mu r$. For $\mu \leq 1$ and $s \geq 0$, $X - Y = (1 - \mu)s \geq 0$, so $\{X \leq x\} \subseteq \{Y \leq x\} \cup \{s < 0\}$ and

$$F(x) \leq G(x) + \Pr(s < 0) = G(x), \quad \Delta(m) = V \int \phi dH_m \geq 0.$$

For $\mu < 1$,

$$\{s > 0\} \subseteq \{Y < X\} = \bigcup_{q \in \mathbb{Q}} \{Y \leq q < X\},$$

so $\Pr(s > 0) > 0$ gives $q \in \mathbb{Q}$ with $\Pr(Y \leq q < X) > 0$, and

$$G(q) \geq \Pr(X \leq q) + \Pr(Y \leq q < X) > F(q).$$

$G - F$ is right-continuous, so $G - F > 0$ on some $[q, u)$.

□

Proof of M3

Lean: EntryContestStep.three_max_sum_real, EntryContestStep.step_identity, EntryContestModel.Delta_step.

Write $K = K_m$, so $H_{m+1} = KF$ and $H_m = KG$. For independent atomless draws $Z \sim K$, $X \sim A$, $Y \sim B$, exactly one is the largest,

$$\int AB dK + \int KB dA + \int KA dB = 1.$$

With $(A, B) = (F, F)$, (G, G) and (F, G) ,

$$\int KF dF = \frac{1}{2} \left(1 - \int F^2 dK \right), \quad \int KG dG = \frac{1}{2} \left(1 - \int G^2 dK \right), \quad \int KG dF + \int KF dG = 1 - \int FG dK,$$

so

$$\frac{\Delta(m+1) - \Delta(m)}{V} = \int KF dF - \int KF dG - \int KG dF + \int KG dG = -\frac{1}{2} \int (F - G)^2 dK.$$

□

Proof of M4

Lean: EntryContestModel.Delta_step_nonpos, EntryContestModel.DeltaUpTo_step, EntryContestEq.Delta_step_strict.

By M3 and $\phi^2 \geq 0$,

$$\Delta(m+1) - \Delta(m) = -\frac{V}{2} \int \phi^2 dK_m \leq 0 \quad (V \geq 0), \quad < 0 \quad \left(V > 0, \int \phi^2 dK_m > 0 \right).$$

For $m \geq Q - 1$ the gain is extended as $\Delta(Q - 1)$, so $\Delta(m+1) \leq \Delta(m)$ for every m .

□

Proof of M5

Lean: EntryContest.single_crossing, EntryContestModel.p5_threshold_from_primitives.

For $i \leq j$, by M4,

$$\kappa_i \leq \kappa_j \leq \Delta(j) \leq \Delta(i).$$

□

Proof of M6

Lean: EntryContestEq.count_invariance_fin, EntryContestMM.exists_mem_ge, EntryContestMM.exists_not_mem_le.

Let $k = |S|$. A k -set of indices has an element $i \geq k - 1$, since $S \subseteq \{0, \dots, k - 2\}$ gives $k \leq k - 1$, and omits some $j \leq k$, since $\{0, \dots, k\} \subseteq S$ gives $k + 1 \leq k$. If $k > k^*$, then $k^* < Q$ and

$$\Delta(k^*) < \kappa_{k^*} \leq \kappa_{k-1} \leq \kappa_i \leq \Delta(k-1) \leq \Delta(k^*).$$

If $k < k^*$, then $j < Q$ and

$$\Delta(k) < \kappa_j \leq \kappa_k \leq \Delta(k).$$

Both are contradictions, so $k = k^*$.

□

Proof of M7

Lean: EntryContest.members_below_kstar, EntryContestEq.assortative_unique.

A member i of an equilibrium has $\kappa_i \leq \Delta(k^* - 1)$ by M6. If $i \geq k^*$,

$$\kappa_i \geq \kappa_{k^*} > \Delta(k^* - 1),$$

so $i < k^*$. Then $S \subseteq \{0, \dots, k^* - 1\}$ and $|S| = k^*$ give $S = \{0, \dots, k^* - 1\}$.

□

Proof of M8

Lean: EntryContestEq.assortative_is_equilibrium, EntryContestEq.assortative_min_cost, EntryContestEq.prize_term_anonymous, EntryContestEq.payoff_gap_anonymous.

(i) For $i < k^*$ and for $k^* \leq j < Q$,

$$\kappa_i \leq \kappa_{k^*-1} \leq \Delta(k^* - 1), \quad \Delta(k^*) < \kappa_{k^*} \leq \kappa_j.$$

(ii) Induct on S , adding its largest element a last. With $T = S \setminus \{a\}$ and $T \subseteq \{0, \dots, a - 1\}$, $|T| \leq a$, so

$$\sum_{i < |T|+1} \kappa_i = \sum_{i < |T|} \kappa_i + \kappa_{|T|} \leq \sum_{i \in T} \kappa_i + \kappa_a = \sum_{i \in S} \kappa_i.$$

(iii) With identical laws, the win probability of i is $P^{\text{in}}(|S|) = \int CF^{|S|-1} G^{Q-|S|} dF$ for $i \in S$ and $P^{\text{out}}(|S|) = \int CF^{|S|} G^{Q-1-|S|} dG$ for $i \notin S$, so with $EU_i(S) = u(w_i) - \kappa_i \mathbf{1}\{i \in S\} + VP_i(S)$,

$$\sum_i EU_i(S) = \sum_i u(w_i) - \sum_{i \in S} \kappa_i + V(|S|P^{\text{in}}(|S|) + (Q - |S|)P^{\text{out}}(|S|)),$$

and the last term is equal for $|S| = |T|$.

□

B Evidence and verification

C Terminology

D Primitives and scope

E Refuted conjectures

F Responses to the referee reports

References